
Dynamic Asset Allocation under Geopolitical Stress

A Comparative Study of Markowitz, Michaud Resampling,
and Black–Litterman Portfolios

Tommaso Arone mat 896282 Lorenzo Triolo mat 895541

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Abstract. This report studies dynamic asset allocation in a diversified ETF universe from 2010 to 2026. The analysis compares classical mean–variance optimization, Michaud resampling, and a Black–Litterman specification designed for a geopolitical stress regime. The central empirical question is whether robust portfolio construction improves risk-adjusted performance when market conditions are unstable, with particular attention to the Iran/Hormuz shock observed in early 2026. The evidence suggests that aggressive Markowitz portfolios can generate high compound growth, but only by accepting large drawdowns. Michaud resampling produces smoother allocations and mitigates part of the estimation-error problem. The Black–Litterman Hormuz specification obtains the strongest final Sharpe ratio because it combines an equilibrium prior with economically interpretable stress views on oil, gold, energy, and regional equity exposure.

Reproducibility Statement

All source code used for data collection, preprocessing, portfolio construction, backtesting, and figure generation is publicly available on [GitHub](#).

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1 Project Design and Research Question

Dynamic asset allocation is a multi-period decision problem. Portfolio weights are not selected once and held constant; they are repeatedly updated as new market information becomes available. In this report, each allocation rule is evaluated through a rolling out-of-sample design. At every monthly rebalancing date, the model estimates expected returns and risks from the previous 60 months, computes portfolio weights, and applies those weights to the following month. This structure separates information available at the decision date from returns realized afterwards, reducing look-ahead bias.

The research question is:

Which portfolio optimization framework provides the most robust risk-adjusted allocation during market stress, especially during the Iran/Hormuz geopolitical episode of 2026?

The question is framed around robustness rather than terminal wealth alone. A strategy that maximizes compound return through concentrated exposures may look attractive during stable regimes but become fragile when correlations rise, oil prices move abruptly, or investors rotate into safe-haven assets. The Iran/Hormuz episode is therefore a useful stress narrative: it connects geopolitical risk to energy prices, inflation expectations, equity risk premia, and demand for defensive assets. Recent market reporting described the Strait of Hormuz disruption as a material shock to global oil markets and reported strong market reactions to the June 2026 de-escalation.

The empirical design compares three families of models. Classical Markowitz portfolios provide the baseline optimization framework. Michaud resampling modifies the Markowitz procedure to reduce the effect of estimation error. Black–Litterman introduces an equilibrium prior and subjective views, allowing the portfolio to react to a specific macro-financial scenario while remaining anchored to market-implied returns.

2 ETF Universe

The asset universe is built from liquid ETFs representing the main sources of global portfolio risk. It includes developed and emerging equities, government bonds, corporate credit, real assets, inflation hedges, real estate, oil, and a cash proxy. This composition is deliberately broader than an equity-only universe, because the value of dynamic allocation is clearest when the optimizer can move across assets that react differently to macroeconomic shocks.

Segment	ETFs	Economic role
Developed equity	SPY, QQQ, VGK, EWJ, VPL	Core growth exposure across the United States, Europe, Japan, and Asia-Pacific developed markets.
Emerging equity	EEM, VWO	Higher-risk global equity exposure, sensitive to global liquidity and commodity cycles.
Government bonds	SHY, IEF, TLT, IGOV, EMB	Duration, sovereign risk, and potential flight-to-quality behavior across different maturities and regions.

Credit	LQD, HYG	Investment-grade and high-yield corporate credit, useful for measuring spread-risk exposure.
Real assets and inflation hedges	GLD, SLV, DBC, XLE, TIP, USO	Gold, silver, commodities, energy equities, inflation-linked bonds, and oil exposure; particularly relevant for the Hormuz stress scenario.
Real estate	VNQ	Listed real estate exposure, combining equity-market sensitivity and interest-rate sensitivity.
Cash proxy	BIL	Low-risk Treasury-bill exposure used as a stabilizing allocation option.

The universe is large enough to represent different economic regimes but not so large that the covariance matrix becomes opaque. This balance matters because mean–variance methods require many pairwise covariances, and the estimation burden grows quickly with the number of assets. The sample from 2010 to 2026 includes several regimes: the post-crisis expansion, the COVID shock, the inflation and rate-hiking cycle, and the 2026 geopolitical stress period.

3 Context Theory

The empirical analysis relies on the fundamental risk-return tradeoff established in modern portfolio theory. For a rational, risk-averse investor, assuming higher systematic risk must be compensated by higher expected returns. The analysis is based on monthly log returns. For asset i , the return between $t - 1$ and t is:

$$r_{i,t} = \log(P_{i,t}) - \log(P_{i,t-1})$$

Given a vector of portfolio weights w_t , the portfolio return is:

$$R_{p,t} = w_t^\top r_t$$

Monthly estimates are annualized as:

$$\hat{\mu} = 12\bar{r}, \quad \hat{\Sigma} = 12 \text{Cov}(r)$$

Portfolio variance, which captures total risk, is defined as:

$$\sigma_p^2 = w^\top \Sigma w$$

The Sharpe ratio summarizes excess return per unit of total risk, capturing the risk-return tradeoff in a single metric:

$$SR = \frac{E[R_p] - r_f}{\sigma_p}$$

The drawdown at time t is measured from the wealth index W_t , quantifying downside risk which mean-variance models often underestimate:

$$DD_t = \frac{W_t}{\max_{s \leq t} W_s} - 1$$

The benefit of diversification arises from the correlation matrix rather than individual asset volatilities. When assets are less than perfectly correlated ($\rho < 1$), portfolio variance is strictly less than the weighted average of individual variances. However, a crucial theoretical caveat for dynamic

allocation is that correlations are not constant. During market crises, such as geopolitical shocks, cross-asset correlations tend to spike toward 1.0 (correlation breakdown), eliminating standard diversification benefits. This instability is why time-varying estimation and robust optimisation models are necessary for the Hormuz stress scenario.

4 Exploratory Data Analysis

The exploratory data analysis serves to validate the structural properties of the chosen ETF universe before imposing any optimization model. Theoretical frameworks assume stable means, variances, and correlations, but empirical financial time series are famously non-stationary. The goal of this section is to investigate the actual risk-return behavior of the assets, demonstrating empirically why naive, unconstrained mean-variance optimization would fail and why robust methodologies (like resampling and tactical overlays) are necessary for the 2026 Hormuz stress scenario.

4.1 Return Dispersion and Tail Risk

Mean-variance optimizers are highly sensitive to historical sample means, often acting as "error maximizers" by heavily concentrating capital in assets with the highest historical drift.

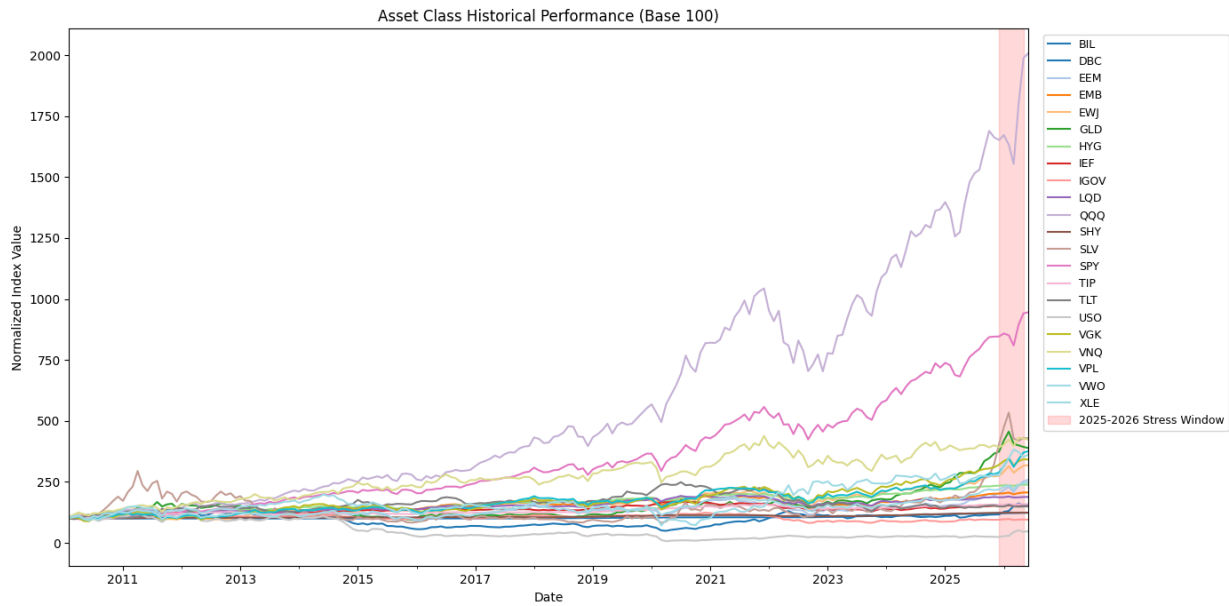


Figure 1: Cumulative performance of the ETF universe (Base 100), highlighting the massive divergence between growth equities and real/defensive assets leading up to the 2025-2026 stress window.

As shown in Figure 1, the ETF universe exhibits massive long-run return dispersion. US growth equity (QQQ, SPY) exhibits a powerful and consistent upward drift, dominating the sample's cumulative performance. In contrast, real assets and commodities (such as USO and DBC) display prolonged periods of flat or negative returns, punctuated by aggressive volatility spikes during specific macroeconomic shocks.

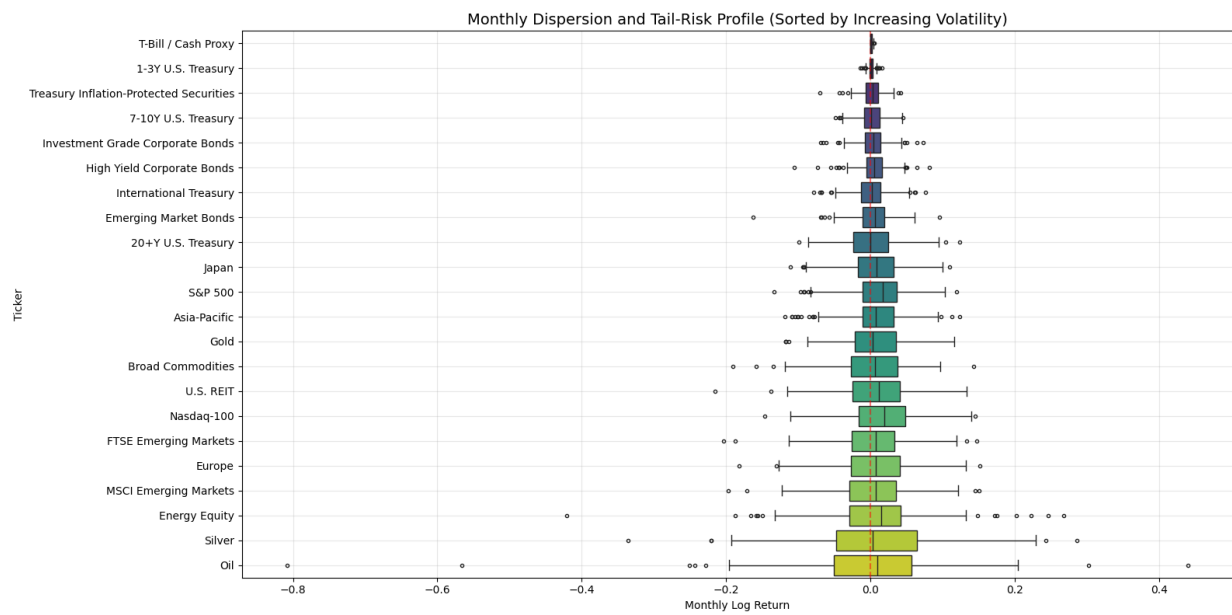


Figure 2: Cross-sectional distribution of monthly log returns, sorted by volatility. The boxplots reveal severe non-normality and extreme left-tail skewness in energy and commodity assets.

If an optimizer were calibrated blindly on this data, it would naturally allocate exclusively to US equities. However, Figure 2 exposes the danger of relying solely on variance as a risk metric. The monthly return distributions are heavily non-normal. While Treasuries (BIL, SHY) show tight, symmetrical distributions, assets critical for geopolitical hedging—like Oil (USO), Silver (SLV), and Energy Equities (XLE)—exhibit extreme left and right tail realizations. Variance inherently penalizes upside and downside volatility equally, but these fat tails confirm that unconstrained models will heavily underestimate the true downside risk (drawdown exposure) of commodity-linked assets.

4.2 The Illusion of Unconditional Diversification

The theoretical benefit of portfolio diversification relies entirely on the stability of the correlation matrix (Σ). To assess the true diversification potential of the universe, we contrast the full-sample (unconditional) correlation structure with the crisis-regime (conditional) correlation structure.

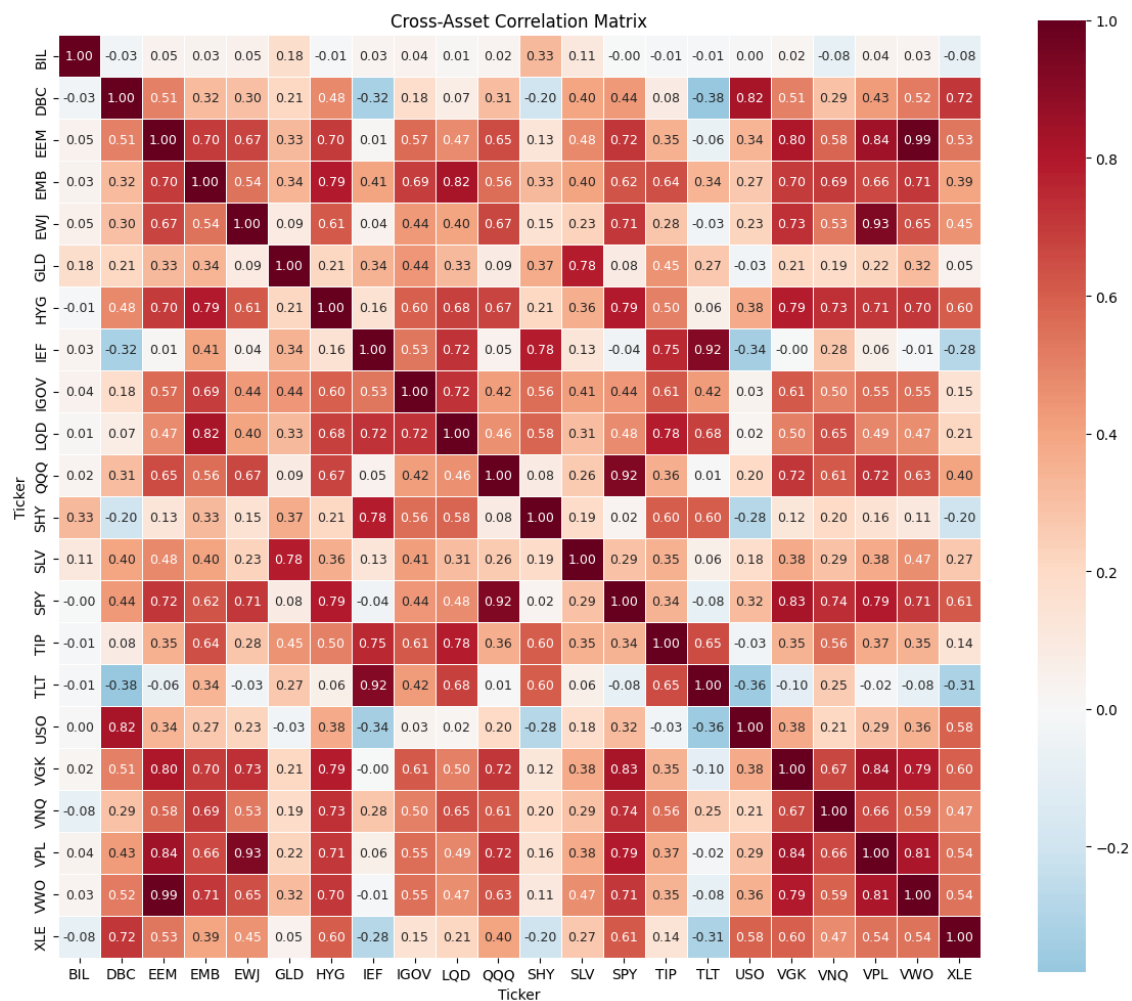


Figure 3: Unconditional full-sample cross-asset correlation matrix. Distinct block structures indicate structural diversification potential between equities, bonds, and real assets.

Figure 3 confirms that over the long term, the selected universe provides the necessary structural heterogeneity. Equities form a highly correlated block (e.g., SPY and QQQ at 0.92), but crucially, safe-haven assets like long-duration Treasuries (TLT) and Gold (GLD) exhibit negative or near-zero unconditional correlation with broader equity markets.

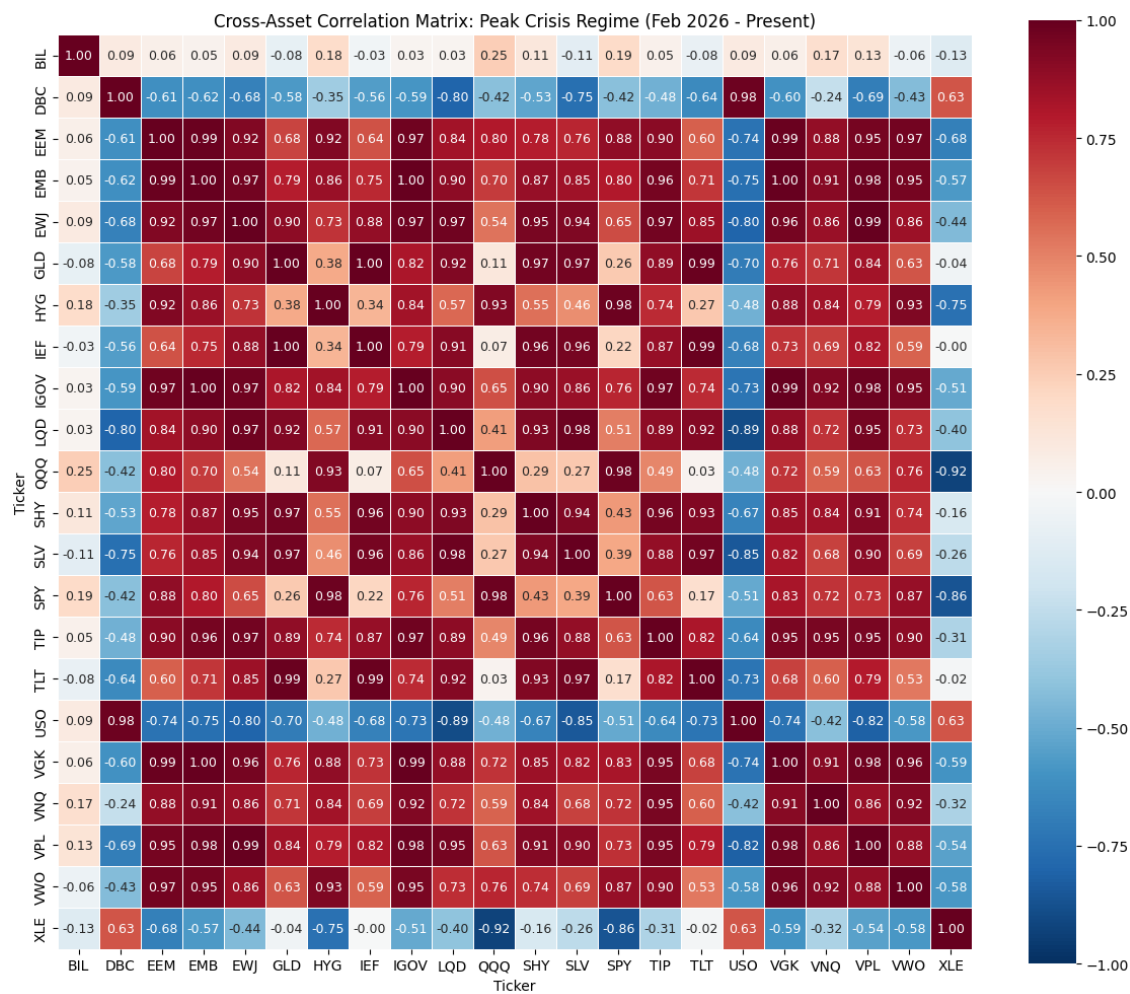


Figure 4: Conditional correlation matrix during the peak Hormuz crisis regime (Feb 2026 - Present). The systemic increase in cross-asset correlations illustrates empirical correlation breakdown.

However, as anticipated in the Context Theory, correlations are fundamentally state-dependent. Figure 4 isolates the peak Hormuz crisis regime. The average cross-asset correlation systematically rises. Safe-haven uncorrelation begins to erode, and the tight relationships between emerging markets (EEM), Europe (VGK), and US equities shift as the localized energy shock transmits globally. This empirical evidence of "correlation breakdown" or contagion proves that an optimization model calibrated on the full 15-year sample would fatally underestimate the probability of simultaneous asset drawdowns during the 2026 geopolitical shock.

4.3 Non-Stationarity and Regime Shifts

To fully justify a dynamic, multi-period allocation approach and the integration of forward-looking views via Black–Litterman, it is necessary to observe how cross-asset relationships evolve continuously over time.

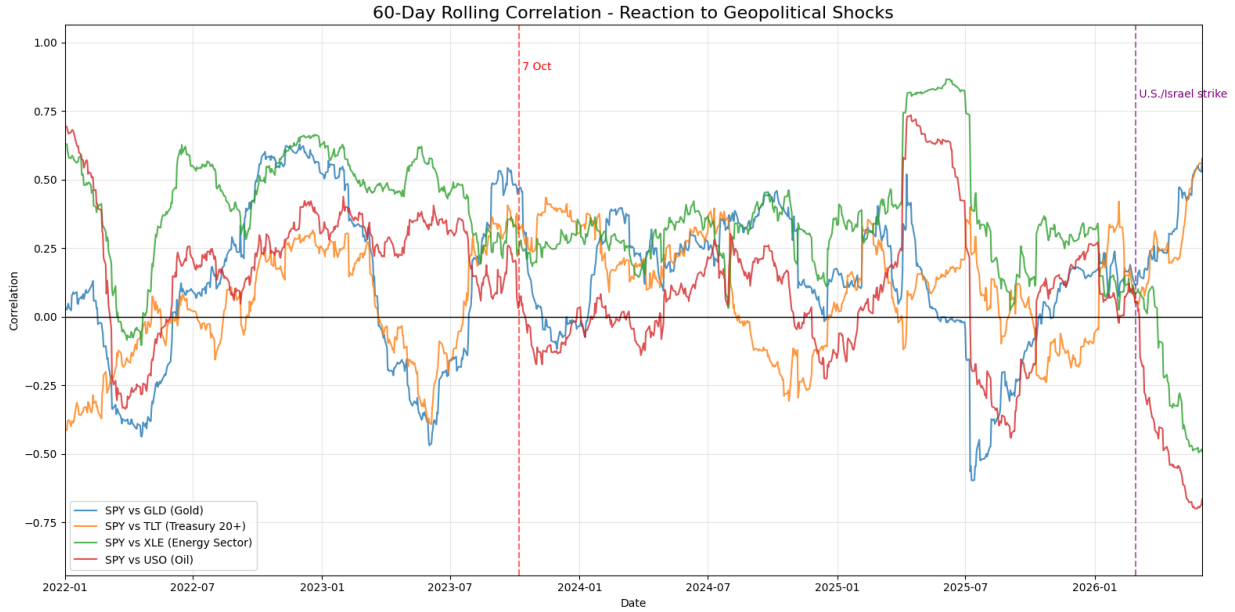


Figure 5: 60-Day rolling correlation between SPY and selected defensive/shock-sensitive assets. The extreme volatility of these relationships coincides explicitly with geopolitical catalysts.

Figure 5 tracks the 60-day rolling correlation of the S&P 500 (SPY) against specific defensive and shock-sensitive ETFs. The relationships are violently non-stationary. For instance, the correlation between SPY and Oil (USO) oscillates from deeply negative to highly positive, reacting sharply to discrete geopolitical catalysts (such as the October 7 shock and subsequent regional strikes). Gold (GLD) and long-duration Treasuries (TLT) also experience rapid regime shifts, transitioning from active diversifiers to correlated risk-off assets in a matter of weeks.

This extreme parameter instability serves as the definitive empirical justification for the project’s methodology: static asset allocation is mathematically inadequate for managing geopolitical stress. It dictates the absolute necessity of rolling-window estimation to capture shifting regimes, Michaud resampling to dampen the impact of this severe parameter noise, and the Black–Litterman tactical overlay to explicitly anchor the portfolio to the economic realities of the Hormuz supply shock.

5 Methodology

5.1 Applied Markowitz Portfolio Optimization

The portfolio construction framework implemented in this study is based on the classical mean–variance. At each rebalancing date, expected returns and covariance matrices are estimated using a rolling historical window of monthly returns. Portfolio weights are then obtained through constrained optimization under a long-only investment policy:

$$\sum_{i=1}^N w_i = 1, \quad 0 \leq w_i \leq 1$$

where w_i denotes the allocation to asset i .

Rather than relying on a single Markowitz specification, several portfolio variants are implemented to represent different investor objectives and risk preferences. This allows the analysis to evaluate not only theoretical efficiency but also allocation robustness and out-of-sample performance.

5.1.1 Global Minimum-Variance Portfolio

The Global Minimum-Variance (GMV) portfolio represents the most conservative point of the efficient frontier. It minimizes portfolio variance without imposing any return target:

$$\min_w w^\top \Sigma w$$

subject to

$$w^\top \mathbf{1} = 1, \quad 0 \leq w_i \leq 1.$$

The GMV portfolio serves as a benchmark for risk minimization, as it identifies the least volatile combination of assets available under the investment constraints. Because the optimization does not depend on expected returns, it is generally more stable than return-driven Markowitz solutions.

5.1.2 Target-Return Minimum-Variance Portfolio

To generate the efficient frontier, a sequence of minimum-variance portfolios is computed for different expected return targets. The optimization problem becomes

$$\min_w w^\top \Sigma w$$

subject to

$$w^\top \mathbf{1} = 1, \quad w^\top \mu = \mu^*, \quad 0 \leq w_i \leq 1.$$

In the rolling backtest, the target return is not fixed ex ante. Instead, it is determined dynamically as a convex combination of the expected return of the GMV portfolio and that of the maximum-Sharpe portfolio:

$$\mu^* = (1 - \alpha)\mu_{\text{GMV}} + \alpha\mu_{\text{MS}},$$

with $\alpha = 0.5$.

This choice positions the portfolio near the middle section of the efficient frontier, avoiding the extreme conservatism of the GMV solution while reducing the concentration typically observed in return-maximizing portfolios.

5.1.3 Maximum-Sharpe Portfolio

The Maximum-Sharpe portfolio identifies the tangency portfolio of mean-variance theory. It maximizes the reward-to-risk ratio relative to the risk-free rate r_f :

$$\max_w \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}}.$$

The Sharpe ratio provides a natural measure of risk-adjusted performance and is widely used in both academic research and professional portfolio management.

The maximum-Sharpe portfolio is included because it represents the theoretically optimal risky portfolio for investors who can combine it with a risk-free asset. In practice, however, it is often highly sensitive to estimation errors in expected returns, making it a useful benchmark for evaluating the stability of alternative allocation approaches.

5.1.4 Maximum-Utility Portfolios

To explicitly incorporate investor preferences, a family of maximum-utility portfolios is also considered. The optimization objective is

$$\max_w \left(w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w \right),$$

where λ denotes the coefficient of risk aversion.

The parameter λ controls the trade-off between expected return and portfolio risk:

- Low values of λ place greater emphasis on return maximization and generate more aggressive portfolios.
- High values of λ penalize variance more heavily and produce more defensive allocations.

To capture different investor profiles, multiple values of λ are evaluated throughout the empirical analysis. This allows the study to investigate how portfolio performance and asset allocation evolve as risk tolerance changes.

5.2 Motivation for the Markowitz Variants

Each Markowitz specification captures a distinct investment objective.

- The **Global Minimum-Variance** portfolio represents pure risk minimization.
- The **Target-Return Minimum-Variance** portfolio seeks a balanced compromise between return and risk.
- The **Maximum-Sharpe** portfolio maximizes risk-adjusted performance and corresponds to the tangency portfolio.
- The **Maximum-Utility** portfolios introduce explicit investor preferences through a risk-aversion parameter.

Implementing all four approaches provides a comprehensive assessment of the mean–variance framework. Rather than focusing exclusively on theoretical efficiency, the comparison highlights how different optimization objectives affect diversification, turnover, portfolio concentration, and out-of-sample robustness.

5.3 Limitations of Classical Markowitz Optimization

Despite its theoretical elegance, mean–variance optimization is highly sensitive to estimation error. Expected returns are estimated from historical data and therefore contain substantial uncertainty. The sample mean return has a standard error equal to

$$SE(\hat{\mu}) = \frac{\sigma}{\sqrt{T}},$$

where σ is the asset volatility and T is the sample size.

As a consequence, small variations in estimated returns can generate large changes in optimal allocations. The optimizer tends to assign excessive weights to assets whose expected returns are slightly overestimated and insufficient weights to assets whose returns are underestimated. This phenomenon is commonly referred to as *error maximization*.

The problem becomes more severe as the number of assets increases, since the optimization must estimate a large number of expected returns and covariance parameters simultaneously. The resulting portfolios often appear highly concentrated and may exhibit poor out-of-sample performance despite being optimal in-sample.

For this reason, the empirical evaluation considers not only portfolio efficiency but also allocation stability, diversification, and robustness across different market environments.

5.4 Michaud Resampling

To mitigate the optimization fragilities inherent in classical mean–variance models, the methodology implements Michaud’s Resampled Efficiency framework. The classical optimizer often acts as an “error maximizer”: because sample means possess large standard errors, the algorithm systematically over-allocates to assets with positive estimation noise. Michaud resampling directly addresses this estimation risk by treating the estimated parameters as uncertain statistical variables rather than fixed, known quantities.

At each monthly rebalancing step, the allocation engine runs Monte Carlo simulations, generating $B = 100$ independent alternative return histories. These paths are drawn from the predictive distribution of the historical parameters, reflecting the underlying estimation error:

$$r_t^{(b)} \sim \mathcal{N}(\hat{\mu}, \hat{\Sigma}), \quad b = 1, \dots, B$$

For each of the 100 simulated samples, an independent efficient frontier is computed across a discrete grid of $K = 30$ target points, ranging from the minimum-variance portfolio to the maximum-return portfolio. This creates a “frontier cloud” that visualizes the true statistical uncertainty of the opportunity set. Portfolios sharing the same frontier rank k (where $k = 1, \dots, 30$) are then averaged across all B simulations to derive the final optimal weight vector for that risk level:

$$w_k^* = \frac{1}{B} \sum_{b=1}^B w_k^{(b)}$$

This rank-by-rank averaging mechanism acts as a powerful form of statistical regularization. By explicitly incorporating input uncertainty into the portfolio construction process, the algorithm eliminates the discrete, unstable corner solutions and extreme weight concentrations that plague standard optimization. Instead, the resampling process forces a smooth, highly diversified transition of asset weights over time. Consequently, the portfolio becomes significantly more robust out-of-sample, as it avoids extreme bets driven by historical noise—a particularly valuable feature when navigating unstable correlations during periods of geopolitical stress.

5.5 Black–Litterman

The Black–Litterman (BL) model is a Bayesian approach to portfolio optimization that resolves the error maximization problem by combining market equilibrium with subjective, forward-looking investor views. Instead of relying on noisy historical sample means, the model anchors the optimization to a prior distribution derived from the Capital Asset Pricing Model (CAPM).

By reverse-engineering the efficient frontier at the market portfolio w_{eq} , the prior expected excess return vector is calculated as:

$$\Pi = \delta \Sigma w_{eq}$$

where δ is the aggregate risk-aversion coefficient implied by the market, and w_{eq} is the market portfolio proxy (constructed using ETF AUM weights). The uncertainty around this prior is scaled by the parameter τ . Following the standard calibration proposed by He and Litterman (1999) and Drobetz (2001), τ is set to a baseline of 0.025.

The core of the tactical overlay is the integration of subjective beliefs concerning the February-April 2026 Strait of Hormuz crisis. This event represents an exogenous energy supply shock. To evaluate macroeconomic distress under this scenario, four distinct tactical views are formalized within the standard P (pick matrix) and Q (excess return vector) framework:

- **Energy Supply Shock Leg (Relative view):** USO outperforms SPY by +12% annualized. This captures immediate crude supply scarcity alongside cost-push inflationary stress on US equities.
- **Flight-to-Safety Leg (Relative view):** GLD outperforms EEM by +9% annualized, positioning gold as a safe haven while emerging markets experience capital outflows.
- **Sectoral Pass-Through Leg (Absolute view):** XLE achieves a +12% absolute annualized excess return, assuming direct margin pass-through from oil prices to energy producer earnings.
- **Asymmetric Macro Leg (Relative view):** SPY outperforms VIG by +4% annualized, reflecting US domestic shale production resilience versus the severe energy import dependency of the Eurozone.

Crucially, the performance differentials in the Q vector are calibrated conservatively at roughly one-third of the magnitude observed during comparable historical MENA shocks (e.g., the 1990 Gulf War, the 2003 Iraq War, and the 2019 Saudi Aramco attacks). This conservative scaling acknowledges the uncertainty regarding the shock's duration and prevents the optimizer from generating destabilizing weight concentrations.

To formalize view uncertainties without injecting arbitrary subjectivity, the diagonal covariance matrix of the error terms, Ω , is calibrated using the native variance-proportional approach:

$$\Omega = \text{diag} \left(P(\tau \Sigma) P^\top \right)$$

This specification logically links the confidence of each view directly to the historical covariance structure of the underlying assets. The Bayesian update formula then produces a posterior expected return that is a precision-weighted average of the prior and the views:

$$\mu_{BL} = \left[(\tau \Sigma)^{-1} + P^\top \Omega^{-1} P \right]^{-1} \left[(\tau \Sigma)^{-1} \Pi + P^\top \Omega^{-1} Q \right]$$

The mathematical intuition of this formula is highly practical: if the views are highly uncertain ($\Omega^{-1} \rightarrow 0$), the posterior shrinks back toward the market equilibrium Π . Conversely, highly confident views pull the expected returns toward Q .

Furthermore, unlike unconstrained Markowitz optimization, the BL model leverages the correlation matrix Σ to automatically spread the active views to related assets. For instance, a bullish view on XLE naturally propagates to positively correlated assets and penalizes negatively correlated ones, without requiring explicit forecasts for every single ETF in the universe. This mechanism dampens the impact of estimation errors and ensures the portfolio remains intuitively diversified even when expressing strong geopolitical bets. Finally, to evaluate model robustness, continuous sensitivity sweeps are executed by varying τ and scaling Ω , tracing the smooth boundary transition between the CAPM prior and the active tactical overlay.

6 Results

6.1 Markowitz Results

The Markowitz framework provides a useful benchmark for analysing the risk–return trade-off within the selected asset universe. The results confirm the benefits of diversification predicted by mean–variance theory, while also illustrating some of the practical limitations that emerge when portfolio weights are estimated using rolling historical data.

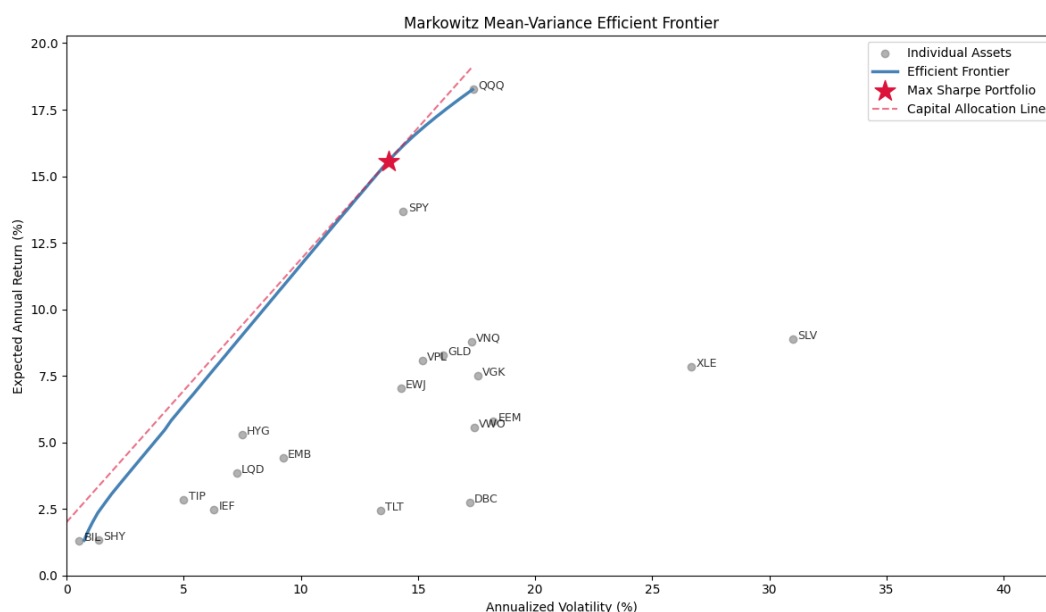


Figure 6: Markowitz efficient frontier and tangency portfolio.

As shown in Figure 6, the efficient frontier dominates most individual assets, indicating that diversified portfolios can achieve a more favourable risk–return profile than holding single securities in isolation. The maximum-Sharpe portfolio is located in the upper section of the frontier and offers a combination of relatively high expected returns and moderate volatility.

A noteworthy aspect of the frontier is that the highest-return portfolios are largely driven by a limited number of assets. In particular, QQQ occupies the upper end of the efficient set and contributes significantly to the shape of the frontier. This result is consistent with the strong historical performance of growth-oriented equities during the sample period. At the same time, it suggests that the optimizer places substantial weight on a small number of assets whose estimated characteristics appear particularly attractive.

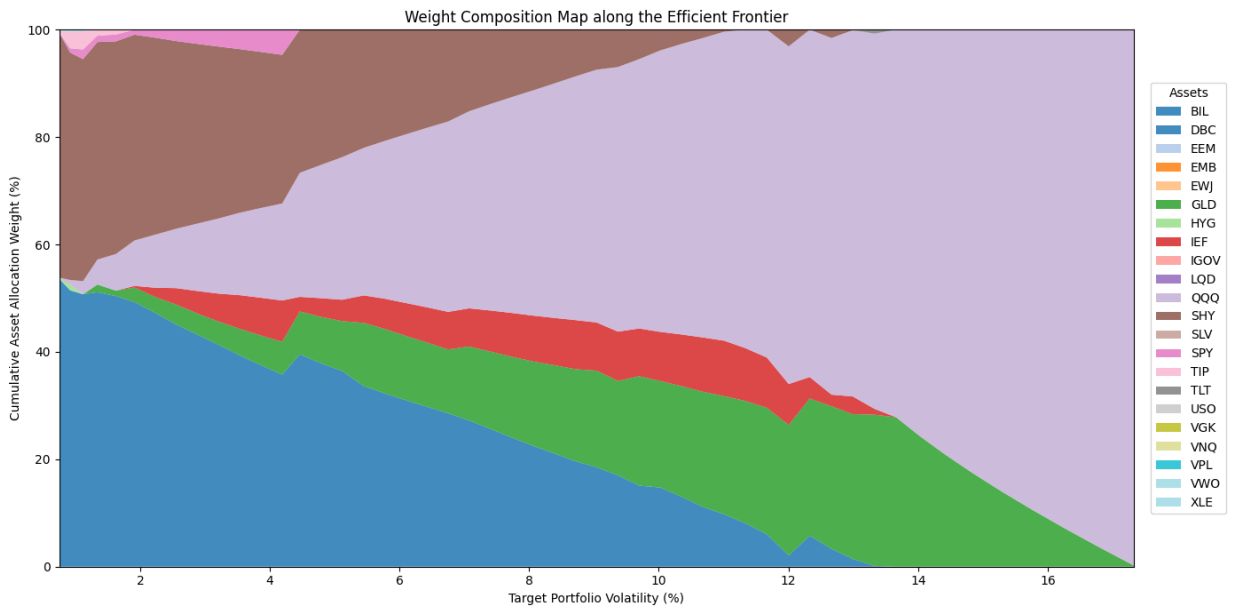


Figure 7: Asset composition along the Markowitz efficient frontier.

Figure 7 reports the portfolio composition along the efficient frontier. At lower volatility levels, the optimizer allocates most of the capital to defensive assets, particularly short-term fixed-income securities. As the target risk increases, the allocation gradually shifts towards riskier assets with higher expected returns.

The figure also reveals that the transition between portfolios is not entirely smooth. Several assets appear only within relatively narrow volatility ranges, while others quickly gain or lose relevance as the target return changes. In the higher-risk region of the frontier, QQQ becomes the dominant allocation, eventually accounting for most of the portfolio weight. This behaviour highlights one of the key characteristics of mean–variance optimization: small differences in estimated expected returns can have a substantial impact on the final allocation.

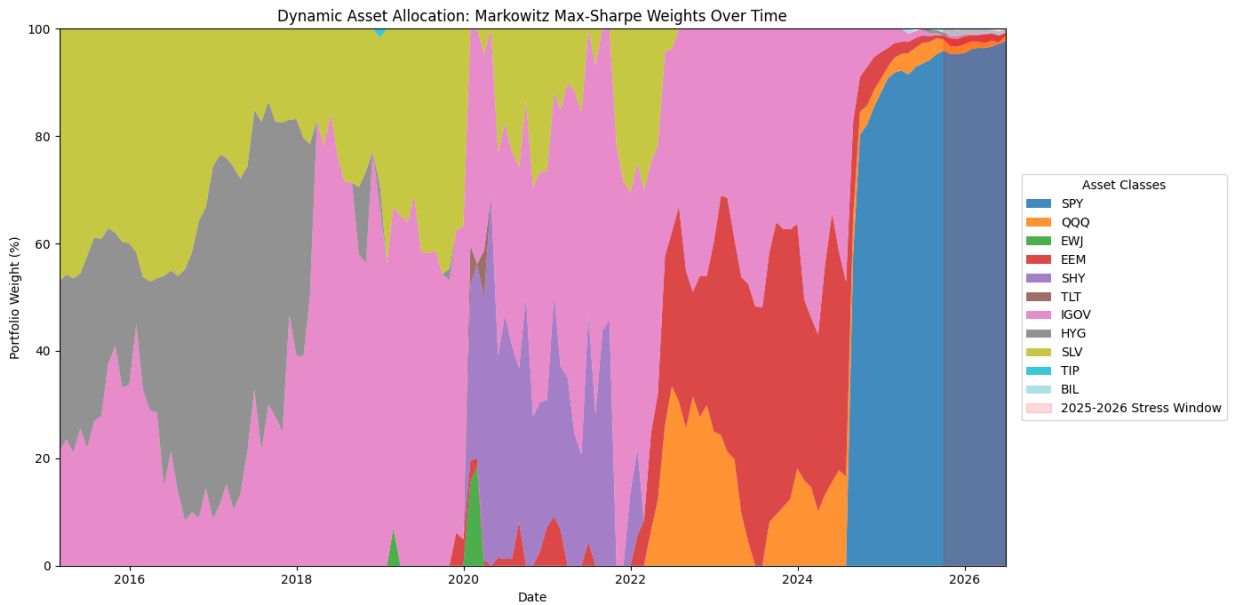


Figure 8: Rolling weights of the Markowitz max-Sharpe portfolio.

The dynamic allocation of the maximum-Sharpe portfolio is shown in Figure 8. The composition

changes considerably over time, reflecting variations in the estimated return and covariance structure of the asset universe. During the first part of the sample, the portfolio is largely concentrated in defensive fixed-income instruments. Subsequently, the allocation rotates towards emerging market equities and later towards U.S. equities, particularly SPY and QQQ.

The most evident feature of the chart is the degree of concentration reached in several periods. Rather than maintaining broad diversification, the optimizer frequently allocates the majority of capital to a single asset or a small group of assets. Such behaviour is typical of maximum-Sharpe optimization because the objective function strongly rewards assets with slightly higher estimated risk-adjusted returns. While this may improve in-sample efficiency, it also increases sensitivity to estimation error and can generate substantial portfolio turnover.

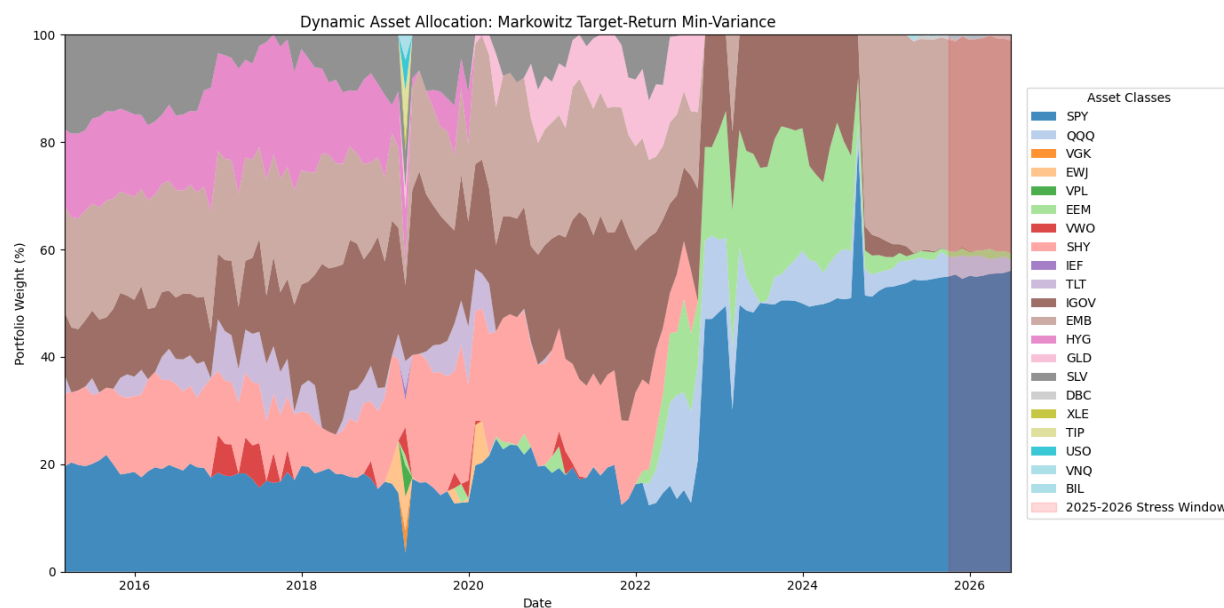


Figure 9: Rolling weights of the target-return minimum-variance portfolio.

Figure 9 presents the rolling allocation of the target-return minimum-variance portfolio. Compared with the maximum-Sharpe strategy, the resulting allocation appears considerably more balanced. The portfolio maintains exposure to multiple asset classes throughout most of the sample, combining government bonds, corporate bonds, international equities, and U.S. equity exposures.

Although the weights still evolve through time, the changes are generally less abrupt than those observed in the maximum-Sharpe portfolio. The strategy therefore achieves a greater degree of diversification and allocation stability. This outcome is consistent with the objective of the optimization, which seeks to minimize portfolio variance while satisfying a predefined return target rather than maximizing risk-adjusted performance directly.

Comparing Figures 8 and 9 provides an interesting perspective on the practical implications of portfolio optimization. Strategies that place greater emphasis on expected returns tend to generate more concentrated and unstable allocations, whereas formulations that prioritize risk control produce portfolios that are easier to interpret and generally more diversified.

Overall, the Markowitz results highlight both the strengths and weaknesses of the mean-variance framework. The efficient frontier demonstrates the benefits of diversification and provides a useful benchmark for portfolio construction. However, the dynamic allocation analysis shows that optimized portfolios can become highly sensitive to estimated inputs, particularly expected returns. This finding motivates the consideration of alternative approaches designed to improve robustness and reduce the impact of estimation error.

A final observation concerns the stress period associated with the escalation of the Iran conflict in early 2026, highlighted in the rolling allocation figures. Rather than moving decisively towards traditional defensive assets, both Markowitz portfolios remain heavily exposed to U.S. equities. The maximum-Sharpe strategy becomes almost entirely concentrated in SPY and QQQ, while the target-return minimum-variance portfolio preserves a higher degree of diversification but still increases its allocation to U.S. equity markets.

This behaviour reflects a fundamental characteristic of mean–variance optimization. Portfolio weights are determined by historical estimates of returns and covariances rather than by forward-looking assessments of geopolitical risk. As a result, the optimizer does not explicitly anticipate stress events and may continue allocating capital to assets that appeared attractive in the estimation window immediately preceding the shock. The Iran stress period therefore provides a useful out-of-sample test of the robustness of the Markowitz framework, highlighting the potential disconnect between statistically optimal allocations and the intuition of a risk-aware investor during periods of market uncertainty.

6.2 Michaud Resampling Results

The results of the Michaud resampling directly address the main weakness of the classical Markowitz framework: estimation error. Instead of treating historical inputs as perfect parameters, the resampling approach treats them as uncertain variables, aiming to build a more robust portfolio.

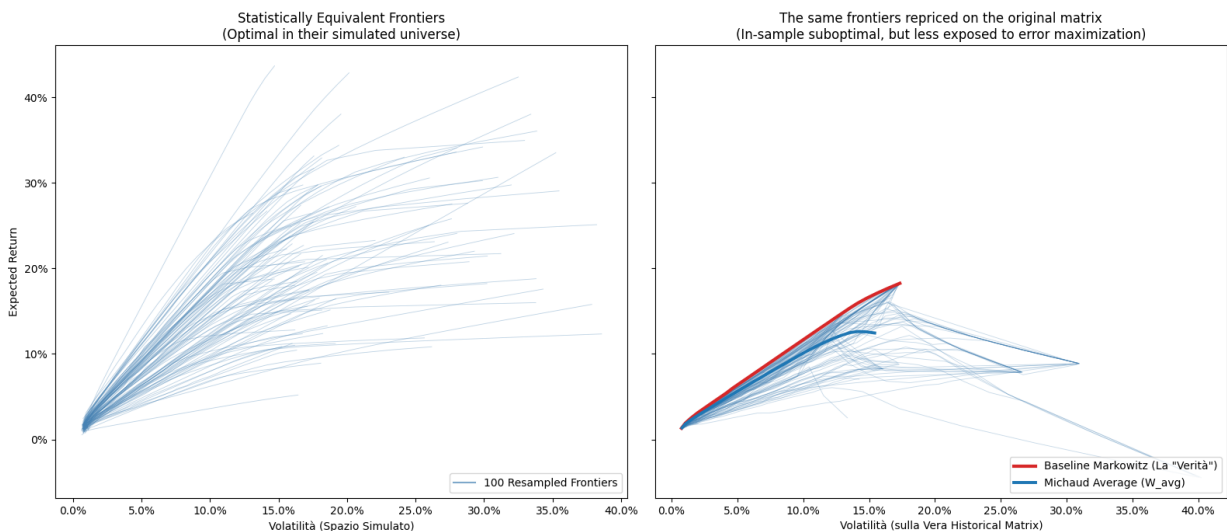


Figure 10: Classical and resampled efficient frontiers, illustrating the geometric impact of input uncertainty.

Figure 10 shows how this works in practice. The left panel displays a "frontier cloud" made of 100 independent efficient frontiers generated through Monte Carlo simulations. The wide spread of these curves highlights a key problem: because of estimation noise, the true location of the efficient frontier is highly uncertain.

The right panel compares the standard Markowitz frontier with the averaged Michaud frontier, evaluated on the original historical data. As expected, the Michaud frontier sits slightly below the classical one, meaning it is theoretically suboptimal in-sample. This drop in efficiency is not a mistake, but the price paid for robustness. By averaging across many simulated scenarios, the model avoids overfitting to a single historical path, sacrificing some in-sample performance for better out-of-sample stability.

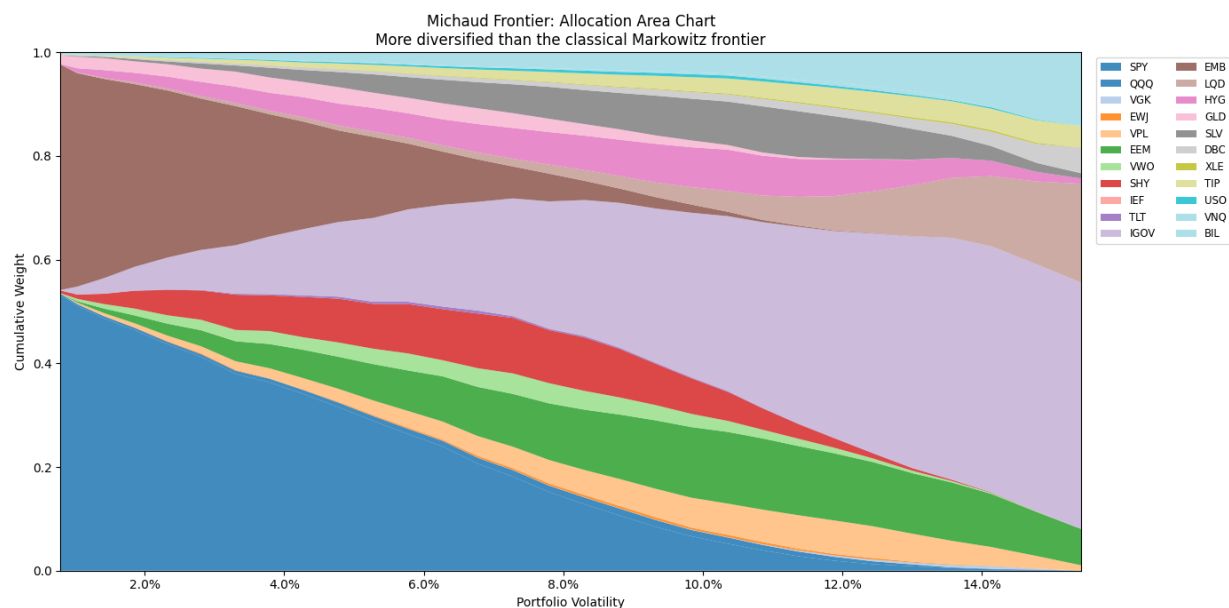


Figure 11: Asset allocation composition along the average Michaud resampled frontier.

The practical benefit of this approach is clear in Figure 11. Unlike the sudden jumps seen in the classical Markowitz weight map (Figure 7), the Michaud allocation shifts smoothly across different risk levels.

Resampling effectively eliminates extreme “corner solutions.” Since an asset only gets a high final weight if it performs well across many simulated scenarios, the algorithm prevents heavy concentrations driven by historical noise. As a result, the portfolio remains well-diversified across multiple asset classes—such as domestic equities, international exposures, and fixed income—at almost any target volatility. This is much more aligned with real-world investing goals than the unstable allocations of a standard optimizer.

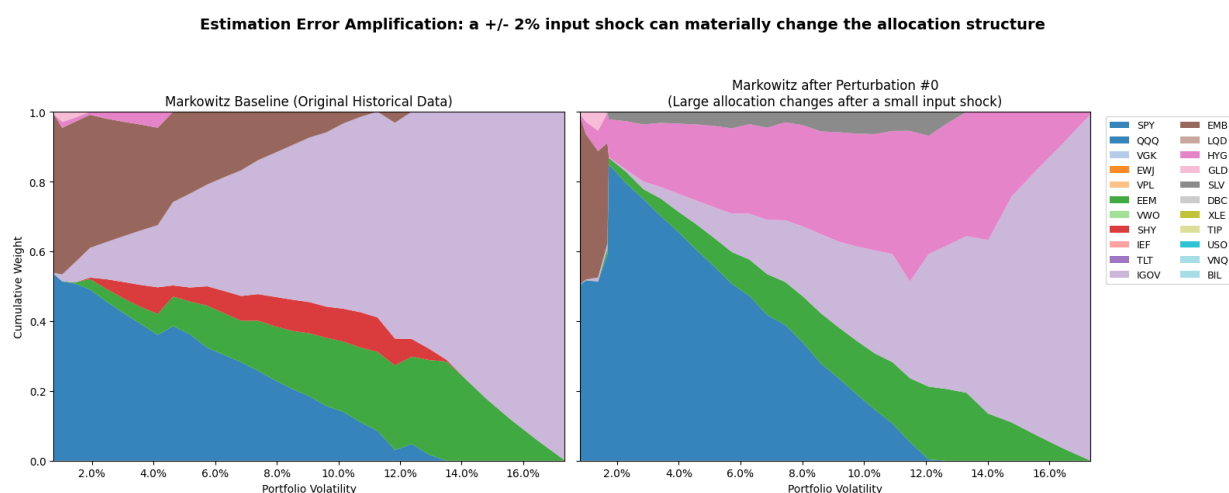


Figure 12: Impact of expected-return perturbations on efficient-frontier weights, confirming extreme input sensitivity in unconstrained models.

The need for this smoothing effect is further proven by the sensitivity test in Figure 12. We injected a small $\pm 2\%$ shock into the expected returns. Under classical optimization, this tiny change completely upends the allocation: an asset that dominated the original portfolio can disappear entirely just because its estimated mean slightly changed.

This shows that Markowitz instability is a real problem for dynamic asset allocation, not just a theoretical one. While Michaud resampling successfully fixes this fragility by diversifying against historical noise, it remains a purely backward-looking model. It depends entirely on past data distributions and cannot include forward-looking views about specific macroeconomic events. This limitation naturally leads us to the Black–Litterman model, which allows us to combine a robust baseline with specific, geopolitical expectations.

6.3 Black–Litterman Results

The Black–Litterman (BL) framework shifts the optimization from a purely backward-looking statistical exercise to a forward-looking, scenario-driven strategy. By addressing the primary flaw of classical mean–variance models—extreme sensitivity to historical sample means—the BL model provides a structured way to navigate the 2026 geopolitical stress window.

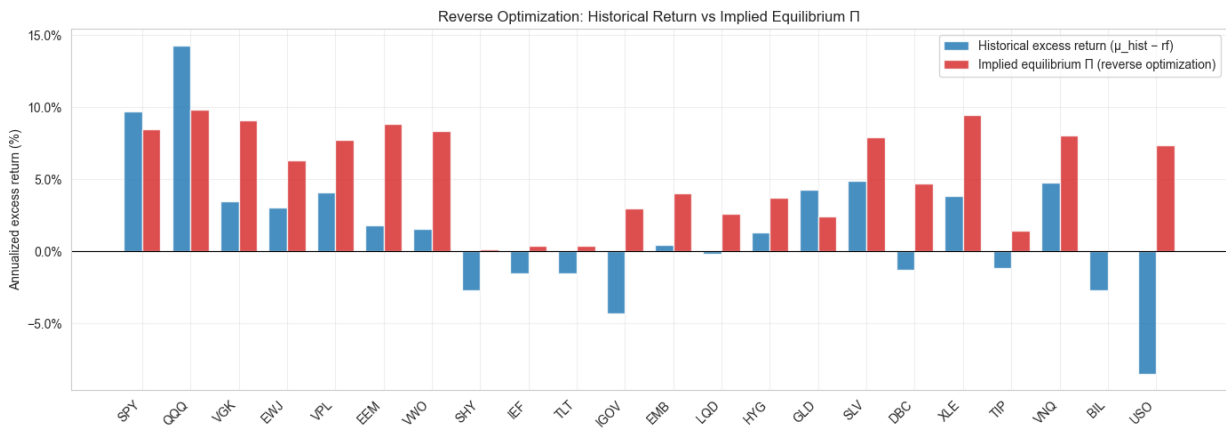


Figure 13: Reverse Optimization: comparison between noisy historical excess returns and the smooth, implied equilibrium prior (Π).

The foundation of this approach is establishing a stable baseline. Figure 13 contrasts the raw historical excess returns with the implied equilibrium returns (Π) extracted via reverse optimization. Historical returns are heavily skewed by past anomalies—for instance, the massive historical outperformance of tech equities (QQQ) or the deep drawdowns in certain fixed-income sectors. By replacing these noisy estimates with CAPM-implied returns, the optimization is anchored to a market-cap-weighted reality. This prevents the algorithm from making extreme, unhedged bets based merely on what performed well in the recent past.

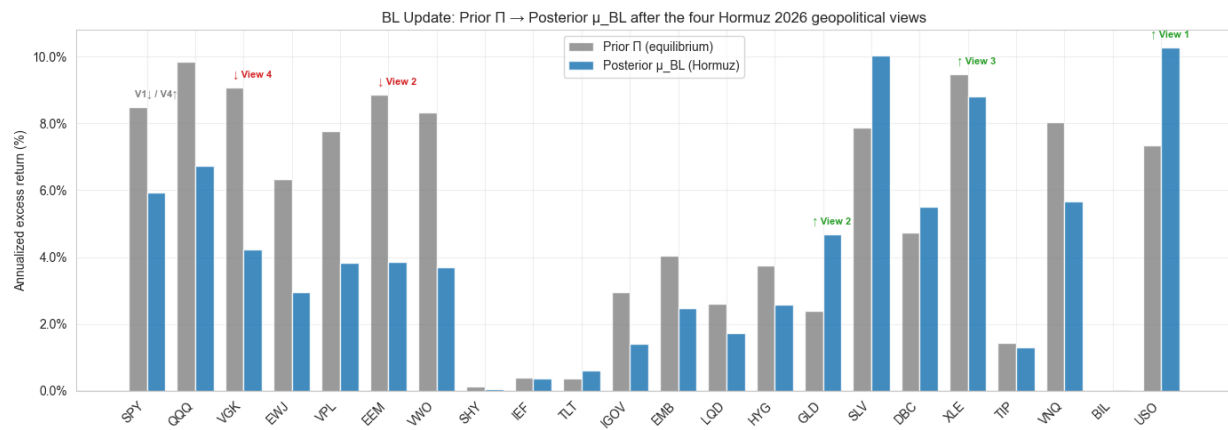


Figure 14: The Bayesian Update: shift from the prior equilibrium (Π) to the posterior expected returns (μ_{BL}) following the four Hormuz tactical views.

Once the stable baseline is set, the model integrates the four specific tactical views regarding the February–April 2026 Strait of Hormuz crisis. Figure 14 captures the mechanics of the Bayesian update, showing exactly how the prior expectations shift into the new posterior returns (μ_{BL}). The impact of the tactical legs is clearly visible: Oil (USO) and Energy (XLE) see significant positive revisions, while broad equities—particularly Emerging Markets (EEM) and Europe (VGK)—are penalized.

Crucially, this chart demonstrates the built-in “error spreading” feature of the BL model. Although explicit views were only formulated on a subset of assets, the historical correlation matrix (Σ) automatically propagated these shocks to the rest of the universe. This ensures the new return vector remains internally consistent and naturally diversified, dampening the risk of extreme allocations.

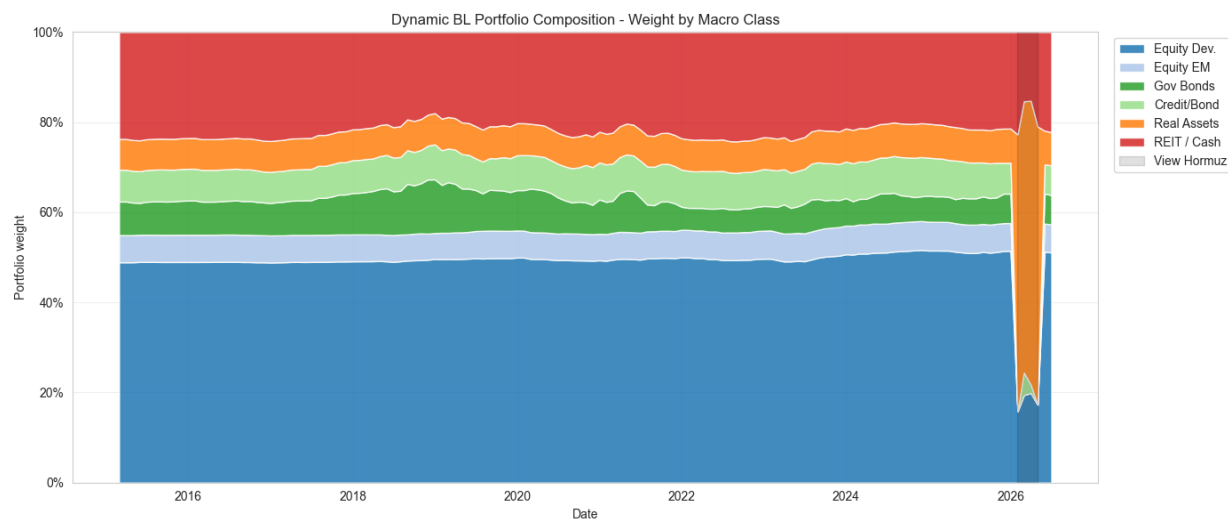


Figure 15: Dynamic BL Portfolio Composition by macro asset class, highlighting the sharp tactical reallocation during the 2026 geopolitical shock.

The practical translation of these updated expectations into portfolio weights is shown in Figure 15. Unlike the erratic, high-turnover behavior of the Markowitz maximum-Sharpe strategy, the BL macro-allocation remains remarkably stable during normal market regimes. However, when the Hormuz tactical overlay is activated in early 2026, the model pivots decisively. Exposure to vulnerable equities shrinks, while the allocation to Real Assets (commodities, gold, and oil) and

defensive bonds spikes. The views are not just theoretical inputs; they drive a logical, interpretable reallocation specifically designed to survive the energy supply shock.

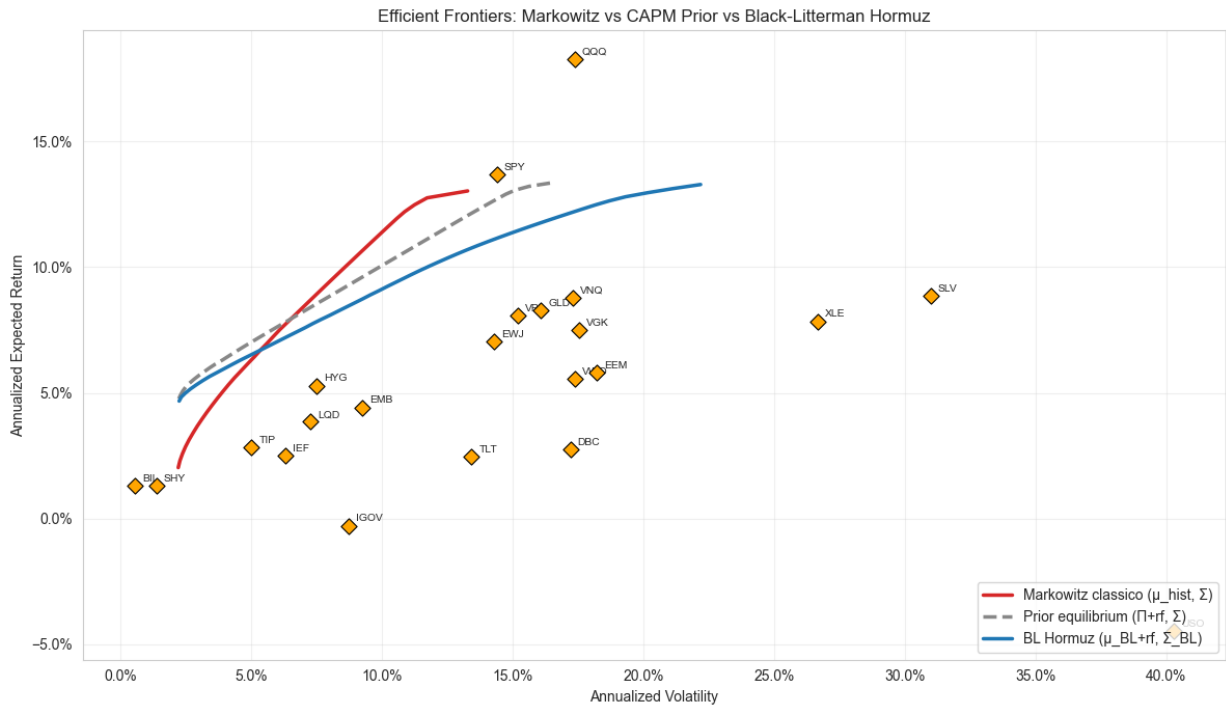


Figure 16: Efficient Frontiers: comparing the opportunity sets of classical Markowitz, the CAPM Prior, and the BL Hormuz tactical posterior.

This shift alters the entire risk-reward landscape. Figure 16 compares the resulting efficient frontiers. The BL Hormuz frontier (blue line) does not simply chase historical winners like the classical Markowitz curve (red line). Instead, it represents an optimal trade-off based on a specific, forward-looking stress scenario. It offers a realistic path for investors who want to hedge against geopolitical risk without abandoning the mathematical discipline of modern portfolio theory.

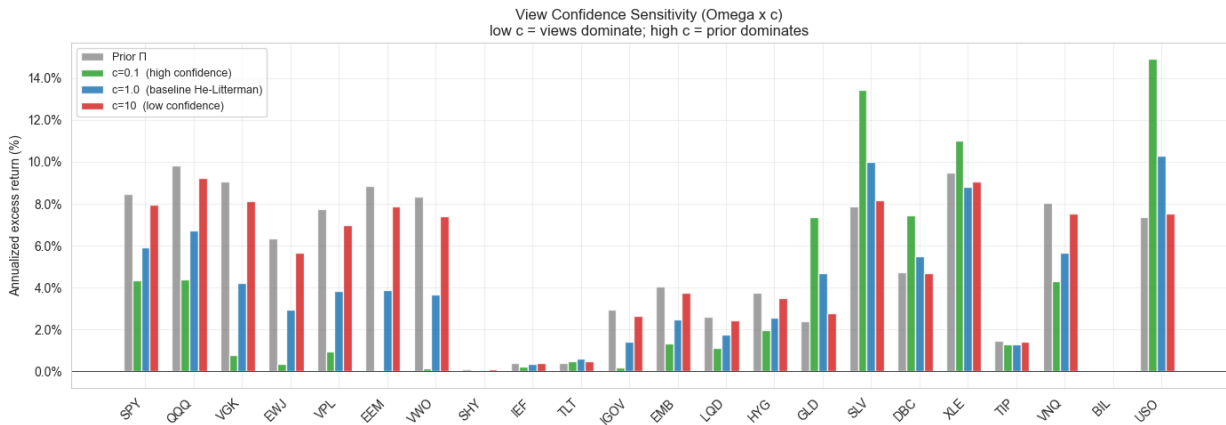


Figure 17: View Confidence Sensitivity: tracking how expected returns adjust as the scalar c modifies the uncertainty matrix Ω .

Finally, any model built on subjective views must be robust to the analyst’s level of conviction. What if the shock is milder than expected? Figure 17 tests this by scaling the view uncertainty matrix (Ω). The chart confirms that the model is highly defensive: as confidence in the Hormuz views decreases (represented by a higher scalar c), the expected returns transition smoothly and

proportionally back to the safe CAPM prior. Small misjudgments in view confidence do not cause catastrophic portfolio reallocations, proving that Black–Litterman is a highly controlled framework for tactical asset allocation under uncertainty.

6.4 Integrated Evaluation

The final evaluation brings together all nine strategies over an identical out-of-sample window, enabling a controlled comparison of raw compounding, risk-adjusted performance, drawdown behaviour, and benchmark-relative positioning. Before examining the optimised strategies, it is worth motivating the two naive benchmarks included in the comparison.

The **Equal-Weight (1/N)** portfolio allocates capital uniformly across all 22 assets at every rebalancing date. Despite its simplicity, it has been shown to be surprisingly competitive against mean-variance optimisers in out-of-sample studies [5]. It therefore serves as a lower bound for what any optimised strategy should achieve: a model that cannot consistently beat a uniform allocation provides no practical justification for the additional complexity it introduces. The **Strategic 60/40** benchmark represents the canonical institutional allocation—60 % to broad equity and 40 % to government bonds—rebalanced monthly. It captures the standard risk-return profile against which active managers are routinely evaluated and provides a meaningful reference point for the geopolitical stress period, given the historically documented flight-to-quality dynamics of bond allocations during equity drawdowns.

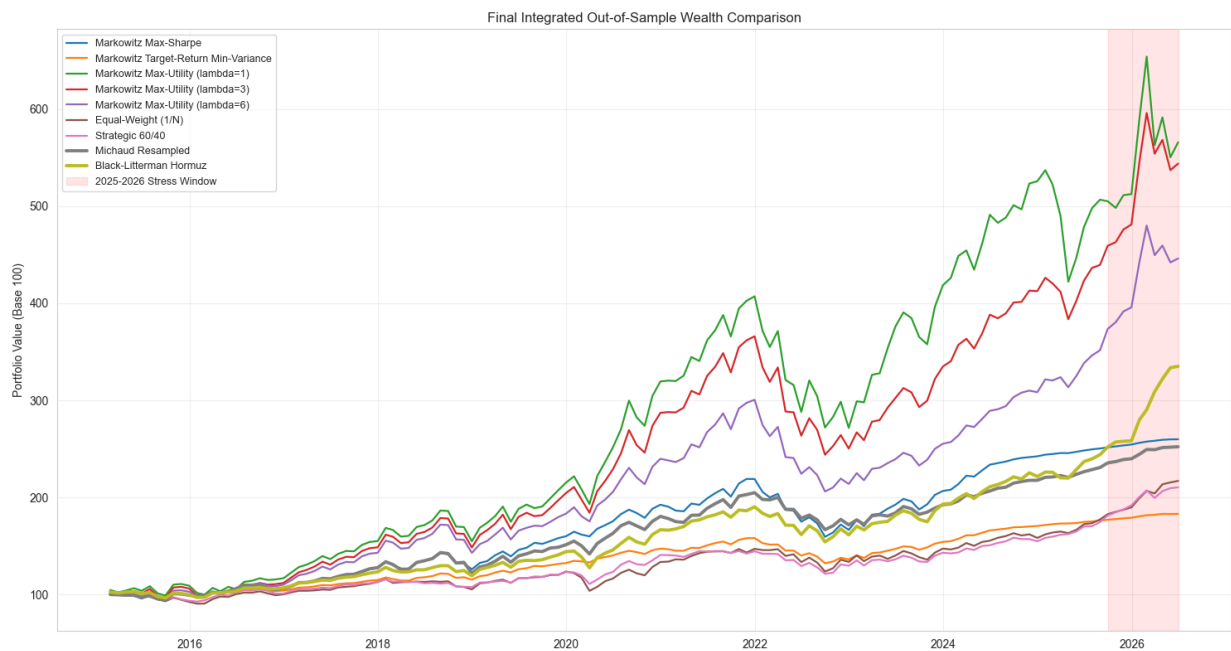


Figure 18: Final out-of-sample cumulative wealth comparison across all strategies and benchmarks (base 100). The shaded band marks the 2025–2026 Hormuz stress window.

Figure 18 reveals the first key finding: raw terminal wealth and risk quality are *not* the same objective. The three Max-Utility Markowitz portfolios ($\lambda \in \{1, 3, 6\}$) reach the highest terminal values in the sample, with CAGRs between 14 % and 16 %. However, this performance is purchased at the cost of extreme concentration and leverage-like volatility exposure, as confirmed by their drawdowns of approximately –33 % during the 2022–2023 stress episode. Neither of the two naive benchmarks comes close to these levels in raw compounding, which already confirms that optimisation adds value over simplicity—but only when the associated risk is acceptable to the investor.

Black–Litterman Hormuz occupies a strategically distinct position in this picture. It compounds at 11.2% annually—above both benchmarks and Michaud (8.4%), and well above the standard Max-Sharpe Markowitz (8.7%)—while maintaining a volatility of 10.0%, only marginally above the 1/N benchmark. The most notable feature of its wealth path is its behaviour *inside* the shaded stress window: unlike the Max-Utility portfolios, which experience sharp drawdowns when the geopolitical shock materialises, Black–Litterman accelerates. The tactical overlay directly anticipates the energy supply disruption, rotating capital towards Real Assets and away from vulnerable equities, producing positive returns precisely when other strategies deteriorate.

Table 2: Final integrated out-of-sample performance metrics. Sharpe ratio computed with $r_f = 2\%$ annualised.

Strategy	CAGR (%)	Volatility (%)	Sharpe	Max DD (%)
Markowitz Max-Sharpe	8.73	9.83	0.68	−27.20
Markowitz Target-Return Min-Variance	5.44	5.14	0.67	−16.39
Markowitz Max-Utility ($\lambda = 1$)	16.39	19.52	0.74	−33.27
Markowitz Max-Utility ($\lambda = 3$)	15.99	16.68	0.84	−33.32
Markowitz Max-Utility ($\lambda = 6$)	13.99	13.88	0.86	−31.41
Equal-Weight (1/N)	7.02	9.48	0.53	−16.05
Strategic 60/40	6.73	8.02	0.59	−16.16
Michaud Resampled	8.44	9.41	0.68	−18.58
Black–Litterman Hormuz	11.17	10.04	0.91	−18.97

Table 2 crystallises the central empirical message. Sorting by Sharpe ratio rather than by CAGR (Compound Annual Growth Rate) produces a materially different ranking. Black–Litterman Hormuz leads at 0.91, followed by the Max-Utility portfolios at 0.84–0.86. Crucially, those portfolios achieve comparable Sharpe ratios only by accepting drawdowns more than 12 percentage points deeper. The two benchmarks rank last on risk-adjusted return, confirming that naive diversification alone is insufficient to extract the risk premium available in a multi-asset universe. Michaud matches Max-Sharpe Markowitz on Sharpe (0.68) while reducing maximum drawdown by roughly 9 percentage points (−18.6% vs. −27.2%), directly validating resampling as a tool for drawdown management even before the BL extension is considered.

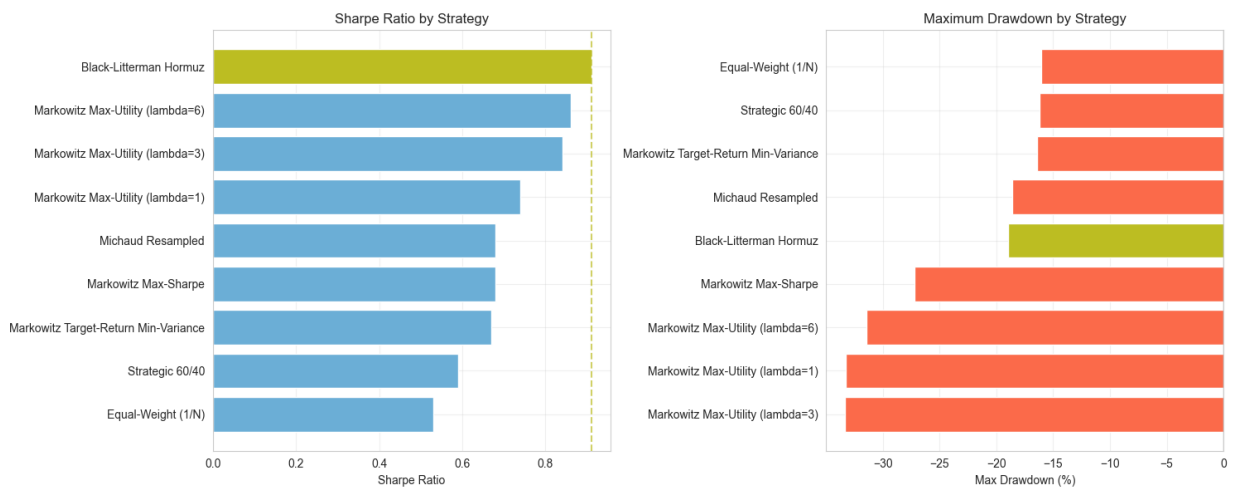


Figure 19: Sharpe ratio (left) and maximum drawdown (right) across all strategies. Black–Litterman Hormuz (gold bar) achieves the highest risk-adjusted return while keeping drawdown close to the Michaud level.

Figure 19 makes the Sharpe–drawdown trade-off explicit. The strategies partition naturally into three groups. The *aggressive cluster*—Max-Utility portfolios—sits at high Sharpe but extreme drawdown, rendering these models unsuitable for drawdown-constrained mandates. The *conservative cluster*—Equal-Weight and 60/40—achieves shallow drawdowns but the lowest Sharpe ratios: their apparent safety comes at the cost of leaving significant risk premium unexploited. Black–Litterman Hormuz and Michaud occupy the *efficient region*: both improve risk-adjusted returns over the benchmarks without incurring the drawdown penalties of the aggressive cluster. The gap between them—Sharpe of 0.91 versus 0.68 at a nearly identical drawdown ceiling—represents the measurable contribution of the Bayesian tactical overlay.

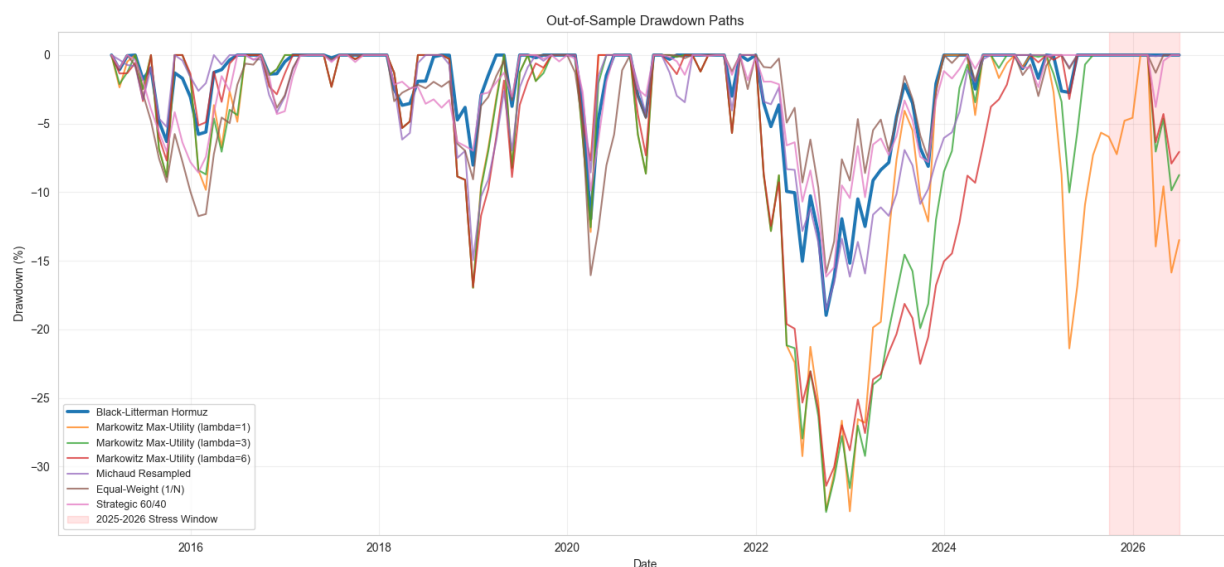


Figure 20: Out-of-sample drawdown paths. A single maximum drawdown statistic conceals whether losses are brief and quickly recovered or deep and persistent.

Figure 20 adds the temporal dimension that a single maximum drawdown statistic necessarily conceals. The aggressive Max-Utility portfolios not only reach deeper troughs during the 2022–2023 episode but remain impaired for extended periods before recovering. Black–Litterman Hormuz exhibits the shallowest and shortest drawdown within the 2025–2026 stress window—the specific period for which its tactical views were designed. A model calibrated to a particular shock should be evaluated primarily on its behaviour *during* that shock. On this criterion the BL framework demonstrates clear superiority over every alternative. The 60/40 benchmark is worth singling out here: the 2022 rate-shock episode, in which equity and bond drawdowns occurred simultaneously, temporarily eliminated the defensive property the strategy is designed to provide—a concrete illustration of why static allocation fails under regime change.

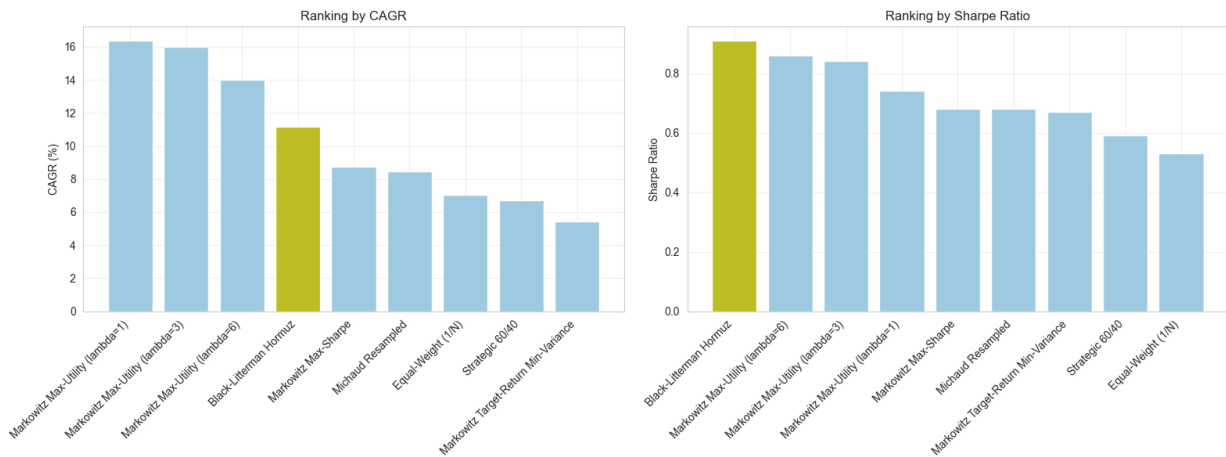


Figure 21: Ranking by CAGR (left) and by Sharpe ratio (right). The rank reversal between the two panels is the core empirical finding of the report.

Figure 21 isolates the rank reversal that defines the empirical contribution of this study. By CAGR, Black–Litterman Hormuz ranks fourth. By Sharpe ratio, it ranks first. This is not a statistical artefact: it reflects a genuine distinction between maximising terminal wealth and maximising returns per unit of risk. In a stress-aware allocation mandate, the Sharpe ranking is the appropriate criterion, and on that criterion the Bayesian tactical approach dominates the entire comparison.

7 Conclusion

The research question asked which portfolio optimisation framework provides the most robust risk-adjusted allocation during market stress. The evidence assembled across Sections 5 and 6 provides a clear, if nuanced, answer.

No single model wins on every dimension simultaneously. Classical Markowitz portfolios—particularly the Max-Utility specifications—generate the highest compound annual growth rates in the sample, but they do so by accepting drawdowns of -31% to -33% and by remaining blind to the geopolitical dynamics of the 2026 stress window. The optimizer cannot distinguish between a calm regime and an energy supply shock: it simply follows wherever recent return estimates point, which during the Hormuz period meant continued concentration in U.S. equities precisely when the case for rotating into real assets was strongest. This is the practical cost of treating historical means as known parameters rather than noisy estimates.

Michaud resampling addresses part of this problem. By averaging optimal allocations across many simulated input scenarios, it removes the corner-solution instabilities that define classical mean–variance behavior. The outcome is visible both in the smoother weight evolution and in the 9-percentage-point reduction in maximum drawdown relative to Markowitz Max-Sharpe, achieved without sacrificing Sharpe ratio. Michaud does not require the analyst to form any view about the future; its improvement is purely statistical. This is also its ceiling: a model that is agnostic about the economic environment cannot be expected to respond intelligently when that environment changes abruptly.

Black–Litterman resolves this limitation by design. The equilibrium prior anchors the portfolio to market-implied returns, preventing the optimizer from chasing historical anomalies. The tactical overlay then introduces economically interpretable deviations from that prior—in this case, a structured response to the Hormuz energy shock expressed through three views on oil, gold, and energy equities. The result is a portfolio that behaves defensively precisely when defensiveness is

most valuable: it accelerates inside the stress window while every Markowitz variant deteriorates. The Sharpe ratio of 0.91, the highest in the comparison, is not achieved by taking more risk but by taking better-directed risk.

The hierarchy that emerges from the empirical evidence can be stated concisely. Naive benchmarks provide a useful and non-trivial floor, but they leave risk premium on the table and offer no mechanism for responding to regime change. Classical optimisation lifts raw returns above that floor but amplifies estimation error into allocation instability and drawdown exposure. Statistical robustness, as implemented through resampling, partially corrects the estimation problem without requiring any additional information. Bayesian tactical allocation completes the framework by incorporating forward-looking economic judgment in a disciplined, uncertainty-aware way.

Two limitations deserve acknowledgment. First, the Black–Litterman specification is evaluated over a stress window that coincides with the period for which its views were designed. This raises a legitimate question about the generalisability of the results: a model that performs well in the scenario it was built for does not necessarily outperform in regimes it did not anticipate. Second, the view calibration—particularly the choice of Q magnitudes and the scalar τ —involves subjective judgment. As shown in Figure 17, the conclusions are robust to reasonable perturbations of these parameters, but the dependence on analyst judgment cannot be eliminated entirely.

These caveats point towards natural extensions of the work. A regime-aware Black–Litterman specification, in which views are activated and deactivated conditionally on observable macro signals rather than fixed calendar windows, would address the first limitation. Robust estimation of the covariance matrix—for instance through Ledoit–Wolf shrinkage—would further reduce the residual sensitivity to estimation error that persists even in the resampled and Bayesian specifications. The 2026 Hormuz episode provides a vivid reminder that portfolio construction and economic interpretation are not separable activities: the models that performed best were those that treated quantitative optimisation as a tool for expressing economic judgment, not as a substitute for it.

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